

Lecture 1. StatML Welcome and Maths Review

COMP90051 Statistical Machine Learning

Lecturer: Jean Honorio



THE UNIVERSITY OF
MELBOURNE

This lecture

- About COMP90051
- Review: Probability theory
- Review: Linear algebra
- Review: Sequences and limits

Subject objectives

- Develop an appreciation for the role of statistical ML, advanced foundations and applications
- Gain an understanding of a representative selection of ML techniques – *how ML works*
- Be able to design, implement and evaluate ML systems
- Become a discerning ML consumer

Subject content

- The subject will cover topics from
Foundations of statistical learning, linear models, non-linear bases, regularised linear regression, generalisation theory, kernel methods, deep neural nets, experts, multi-armed bandits, Bayesian learning, probabilistic graphical models
- Classical / in-person teaching
 - * Theory in lectures (2x per week), subject assessment;
 - * Hands-on experience in weekly workshop (starts week 2), subject assessment;
 - * *Strongly recommend you keep up (viz. quizzes)*

Subject staff / Contact

Lecturers: [Jean Honorio](#) [Lectures 1-10, 15, 20-21]
Zahra Dasht Bozorgi [Lectures 11-14, 16-19, 22-24]

Tutors: Afsaneh Hasanebrahimi (Head tutor)
Yiyi Guo, Jinhao Li, Jedwin Villanueva
See Canvas for latest list, contact details, tutorial location/time.

Ed discussion *Will open after this first lecture*

Contacting staff *Combined staff email*
comp90051-2026s1-staff@lists.unimelb.edu.au

Advanced ML: Expected Background

- **ML:** COMP90049, COMP20008, MAST90105
- **Algorithms & complexity:** big-O, termination; basic data structures & algorithms
- **Solid coding**, ideally experience in Python
- **Mathematics and statistics**

Maths / stats requirements

- **Probability theory**: probability calculus; discrete/continuous distributions; multivariate; exponential families; Bayes rule
- **Sequences**: sequences, limits, supremum
- **Linear algebra**: vector inner products & norms; orthonormal bases; matrix operations, inverses, eigenvectors/values
- **Calculus & optimisation**: partial derivatives; gradient descent; convexity; Lagrange multipliers

Some statistics about students in this class

52% Master Data Science

42% Master Information Technology

2% Master Software Engineering

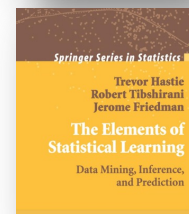
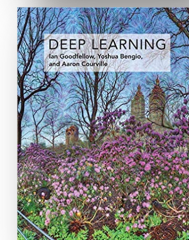
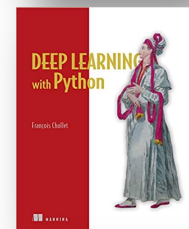
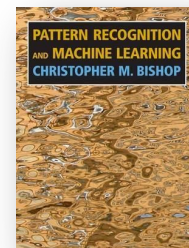
4% other Master programs

Some know a bit more programming than others

Some know a bit more math than others

Textbooks

- We **don't have only one reference**. We prefer to pick good bits from several. We may also supplement with other readings as we go.
- All are available free online or through the library digitally. See the **Canvas subject overview** for links. Therefore, **no need to buy**.
- Primarily we refer to (good all rounder): ***Bishop (2007) Pattern Recognition and Machine Learning***
- ***Practical Deep Nets: Chollet (2017) Deep learning with Python***
- ***More deep learning detail: Goodfellow, Bengio, Courville (2016) Deep learning***
- For more on PGMs/Bayesian inference: Murphy (2012) *Machine Learning: A Probabilistic Perspective*
- For reference on frequentist ideas, SVMs, lasso, etc.: Hastie, Tibshirani, Friedman (2001) *The Elements of Statistical Learning: Data Mining, Inference and Prediction*
- For reference on learning theory (though goes deeper/broader than we need) Shalev-Shwartz and Ben-David "*Understanding Machine Learning: From Theory to Algorithms*"

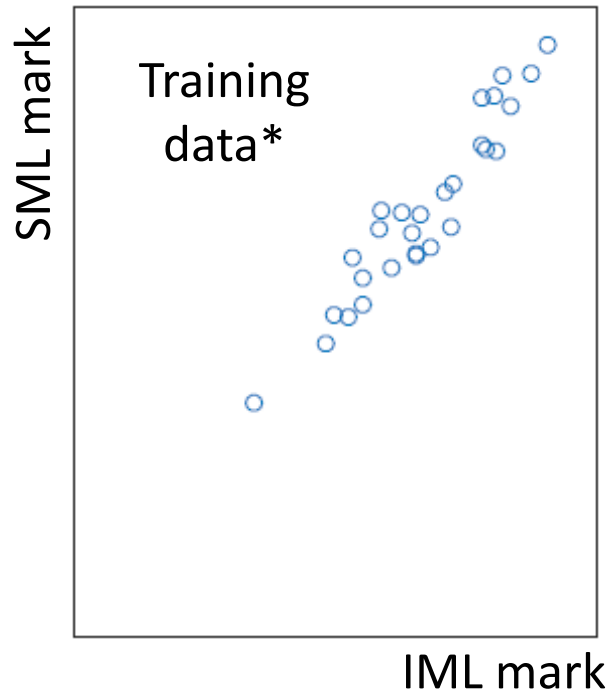


Assessment

- Assessment components
 - * Timing may differ from handbook due to non-teaching week
 - * An individual project: from week 4, due week 7 (20%, individual)
 - *Late submissions incur 10% penalty per day or fraction. Late submissions after 3 days not allowed.*
 - * A group project: from week 8, due week 11 (30%, group size 3)
 - *Late submissions incur 10% penalty per day or fraction. Late submissions after 3 days not allowed.*
 - * Quizzes 30min fortnightly (10% - best 5 of 6)
 - Q1 to **open March 15 at 11.59pm** and **close March 16 at 11.59pm**
 - *Late submissions not allowed.*
 - * Final Exam (40%): 2hr closed book on-campus exam (40%)
- 50% hurdles applied to **exam**

Probability theory

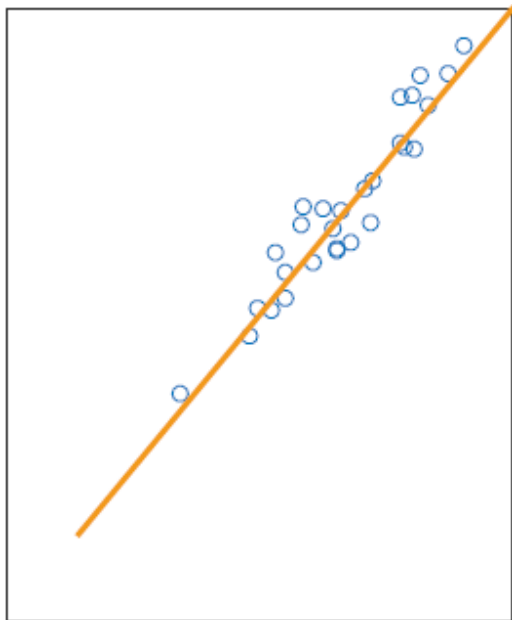
Data is noisy (almost always)



- Example:
 - * given mark for Intro ML (IML)
 - * predict mark for Stat Machine Learning (SML)

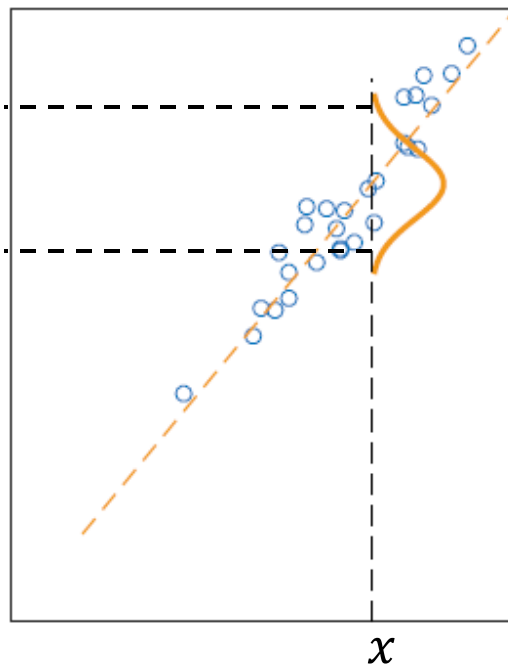
* synthetic data :)

Types of models



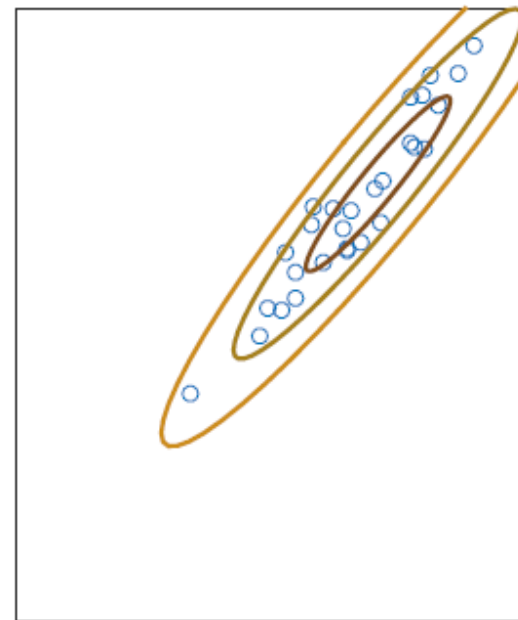
$$\hat{y} = f(x)$$

IntroML mark was 95,
SML mark is predicted
to be 95



$$P(y|x)$$

IntroML mark was 95,
SML mark is likely to
be in (92, 97)



$$P(x, y)$$

probability of having
($IML = x, SML = y$)

Basics of probability theory



- A probability space:
 - * Set Ω of possible outcomes
 - * Set F of events (subsets of outcomes)
 - * Probability measure $P: F \rightarrow \mathbb{R}$
- Example: a die roll
 - * $\{1, 2, 3, 4, 5, 6\}$
 - * $\{\varnothing, \{1\}, \dots, \{6\}, \{1,2\}, \dots, \{5,6\}, \dots, \{1,2,3,4,5,6\}\}$
 - * $P(\varnothing)=0$, $P(\{1\})=1/6$,
 $P(\{1,2\})=1/3$, ...

Axioms of probability

F contains all of: Ω ; all complements $\Omega \setminus f$, $f \in F$; the union of any countable set of events in F .

1. $P(f) \geq 0$ for every event $f \in F$.
2. $P(\cup_f f) = \sum_f P(f)$ for all countable sets of pairwise disjoint events.
3. $P(\Omega) = 1$

Random variables (r.v.'s)



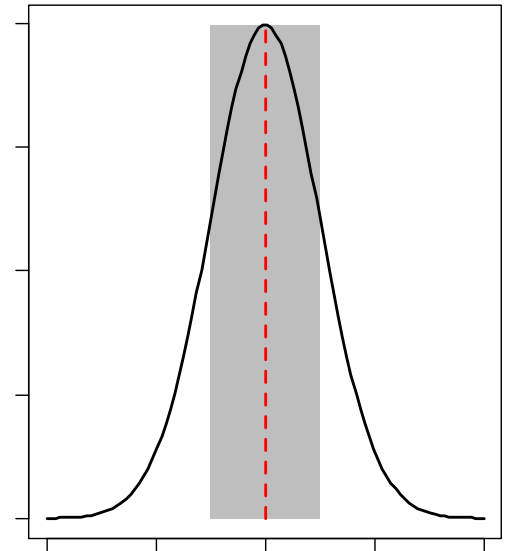
- A random variable X is a numeric function of outcome $X(\omega) \in \mathbf{R}$
- $P(X \in A)$ denotes the probability of the outcome being such that X falls in the range A
- Example: X winnings on \$5 bet on even die roll
 - * X maps 1,3,5 to -5
 - X maps 2,4,6 to 5
 - * $P(X=5) = P(X=-5) = \frac{1}{2}$

Discrete vs. continuous distributions

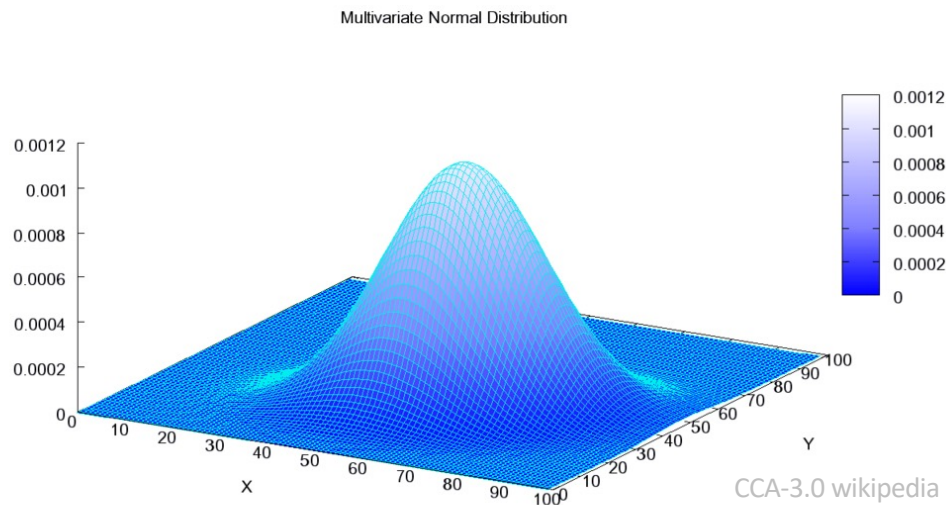
- Discrete distributions
 - * Govern r.v. taking discrete values
 - * Described by **probability mass function** $p(x)$ which is $P(X=x)$
 - * $P(X \leq x) = \sum_{a=-\infty}^x p(a)$
 - * **Examples:** Bernoulli, Binomial, Multinomial, Poisson
- Continuous distributions
 - * Govern real-valued r.v.
 - * Cannot talk about PMF but rather **probability density function** $p(x)$
 - * $P(X \leq x) = \int_{-\infty}^x p(a) da$
 - * **Examples:** Uniform, Normal, Laplace, Gamma, Beta, Dirichlet

Expectation

- Expectation $E[X]$ is the r.v. X 's “average” value
 - * Discrete: $E[X] = \sum_x x P(X = x)$
 - * Continuous: $E[X] = \int_x x p(x) dx$
- Properties
 - * Linear: $E[aX + b] = aE[X] + b$
 $E[X + Y] = E[X] + E[Y]$
 - * Monotone: $X \geq Y \Rightarrow E[X] \geq E[Y]$
- Variance: $Var(X) = E[(X - E[X])^2]$



Multivariate distributions

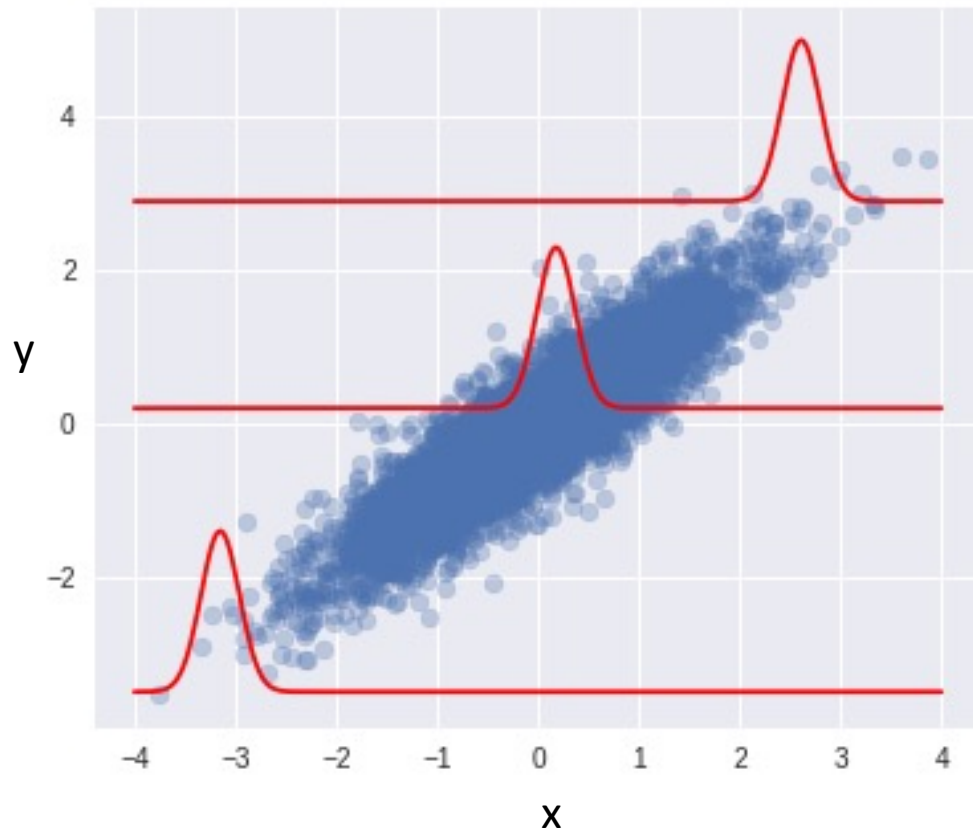


- Specify joint distribution over multiple variables
- Probabilities are computed as in univariate case, we now just have repeated summations or repeated integrals
- Discrete: $P((X, Y) \in A) = \sum_{(x,y) \in A} p(x, y)$
- Continuous: $P((X, Y) \in A) = \int_A p(x, y) dx dy$

Independence and conditioning

- X, Y are **independent** if
 - * $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$
 - * Similarly for densities:
 $p_{X,Y}(x, y) = p_X(x)p_Y(y)$
 - * **Intuitively**: knowing value of Y reveals nothing about X
 - * **Algebraically**: the joint on X, Y factorises!
- **Conditional probability**
 - * $P(A|B) = \frac{P(A \cap B)}{P(B)}$
 - * Similarly for densities
 $p(y|x) = \frac{p(x,y)}{p(x)}$
 - * **Intuitively**: probability event A will occur given we know event B has occurred
 - * **The Product Rule**:
 $P(A \cap B) = P(A|B) P(B)$
 - * X, Y independent equiv to
 $P(Y = y|X = x) = P(Y = y)$

Independence and conditioning



- Not independent: knowing the value of y gives you insight into the value that x will take

Inverting conditioning: Bayes' Theorem

- In terms of events A, B

- * $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

- * $P(A|B) = \frac{P(B|A) P(A)}{P(B)}$



Bayes

- Simple rule that lets us swap conditioning order

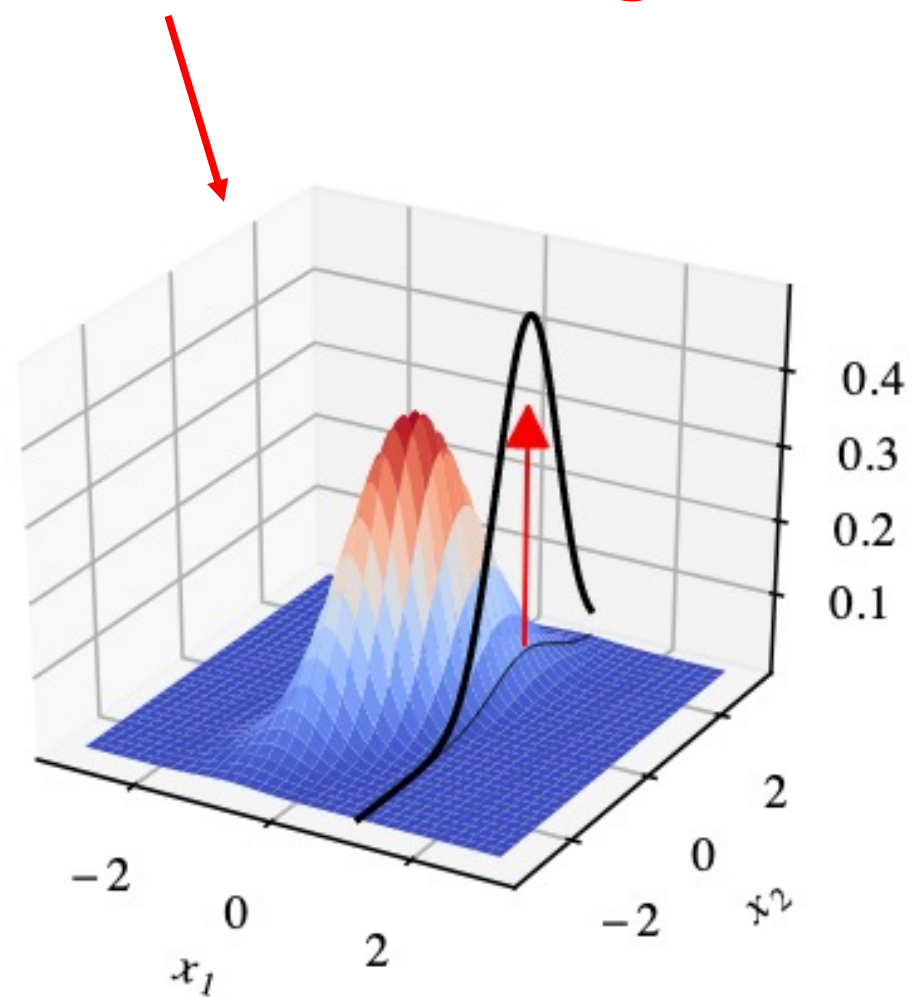
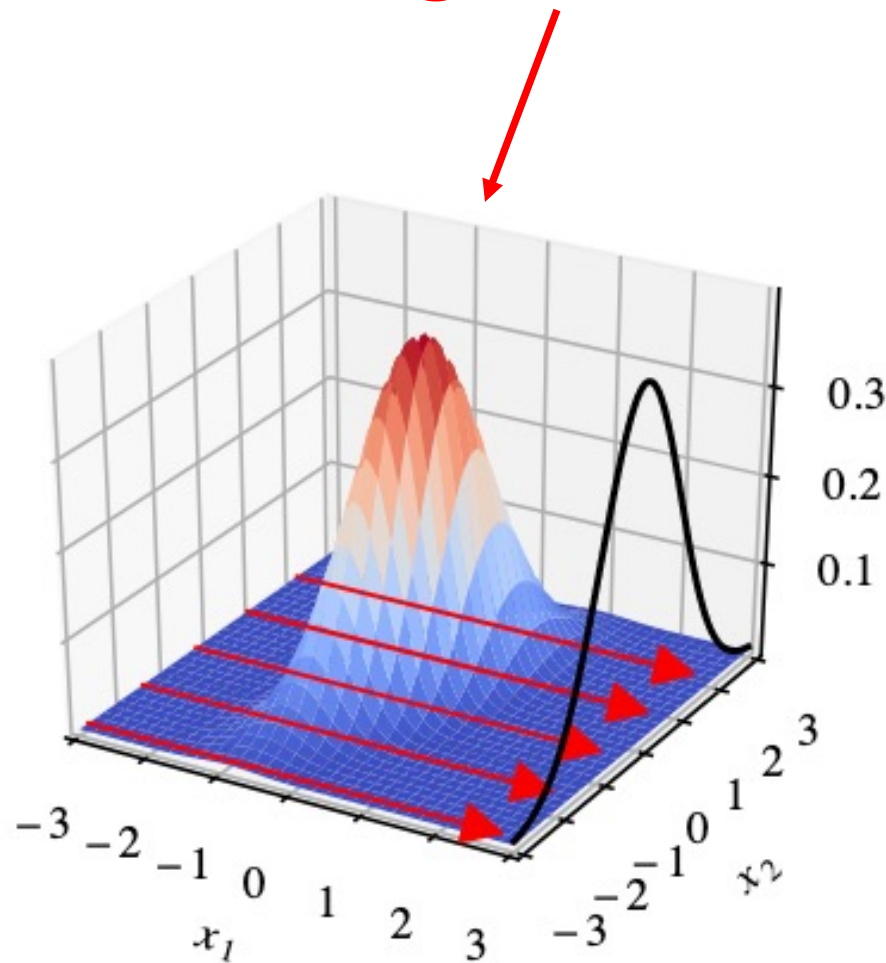
Inverting conditioning: Bayes' Theorem



Bayes

- In terms of events A, B
 - * $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$
 - * $P(A|B) = \frac{P(B|A) P(A)}{P(B)}$
- Simple rule that lets us swap conditioning order
- Probabilistic and Bayesian inference make heavy use
 - * **Marginals**: probabilities of individual variables
 - * **The Sum Rule** of marginalisation: summing away all but r.v.'s of interest. $P(A) = \sum_b P(A, B = b)$

Marginalisation vs conditioning



Mini Summary

- Probability spaces, axioms of probability
- Discrete vs continuous; Univariate vs multivariate
- Expectation, Variance
- Independence and conditioning
- Bayes rule and marginalisation

Next: Linear algebra primer/review

Vectors

Link between geometric and algebraic
interpretation of ML methods

What are vectors?

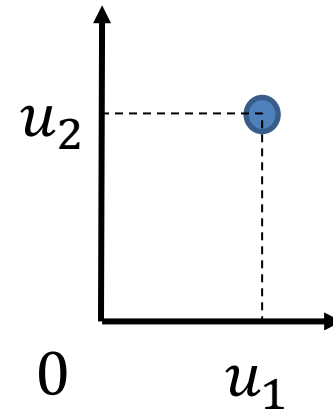
Suppose $\mathbf{u} = [u_1, u_2]'$. What does $\mathbf{u}$ really represent?



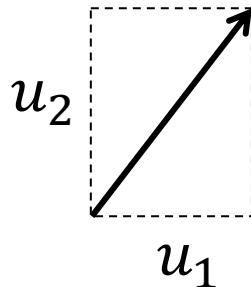
Ordered set of numbers $\{u_1, u_2\}$



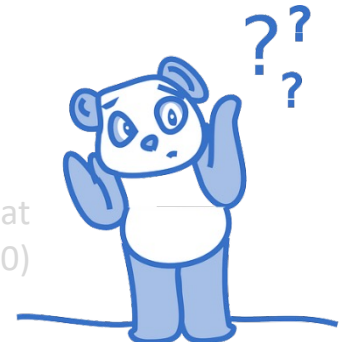
Cartesian coordinates of a point



A direction



art: OpenClipartVectors at
pixabay.com (CC0)

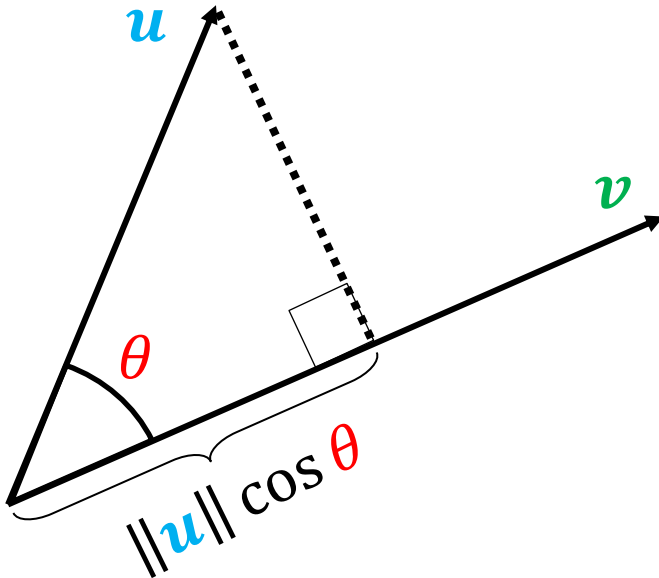


Dot product: Algebraic definition

- Given two m -dimensional vectors $\mathbf{u}$ and $\mathbf{v}$, their dot product is $\mathbf{u} \cdot \mathbf{v} \equiv \mathbf{u}'\mathbf{v} \equiv \sum_{i=1}^m u_i v_i$
 - * E.g., weighted sum of terms is a dot product $\mathbf{x}'\mathbf{w}$
- If k is a scalar, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are vectors then
$$(k\mathbf{a})'\mathbf{b} = k(\mathbf{a}'\mathbf{b}) = \mathbf{a}'(k\mathbf{b})$$
$$\mathbf{a}'(\mathbf{b} + \mathbf{c}) = \mathbf{a}'\mathbf{b} + \mathbf{a}'\mathbf{c}$$

Dot product: Geometric definition

- Given two m -dimensional Euclidean vectors $\mathbf{u}$ and $\mathbf{v}$, their dot product is $\mathbf{u} \cdot \mathbf{v} \equiv \mathbf{u}'\mathbf{v} \equiv \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$
 - * $\|\mathbf{u}\|, \|\mathbf{v}\|$ are L_2 norms for $\mathbf{u}, \mathbf{v}$ also written as $\|\mathbf{u}\|_2$
 - * θ is the angle between the vectors



The *scalar projection* of $\mathbf{u}$ onto $\mathbf{v}$ is given by

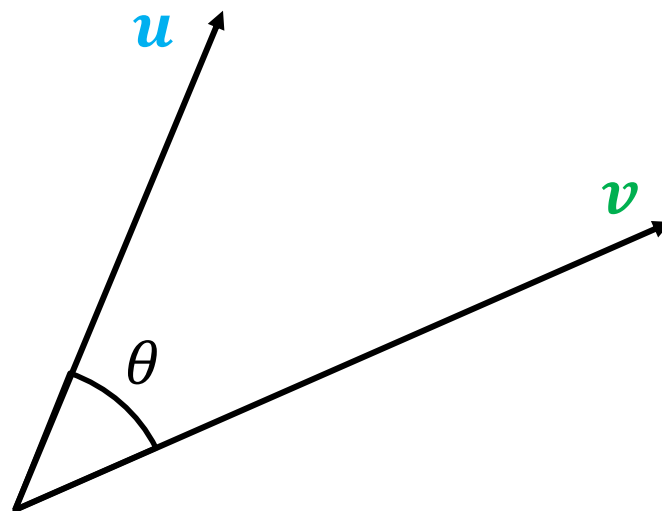
$$u_v = \|\mathbf{u}\| \cos \theta$$

Thus dot product is

$$\mathbf{u}'\mathbf{v} = u_v \|\mathbf{v}\| = v_u \|\mathbf{u}\|$$

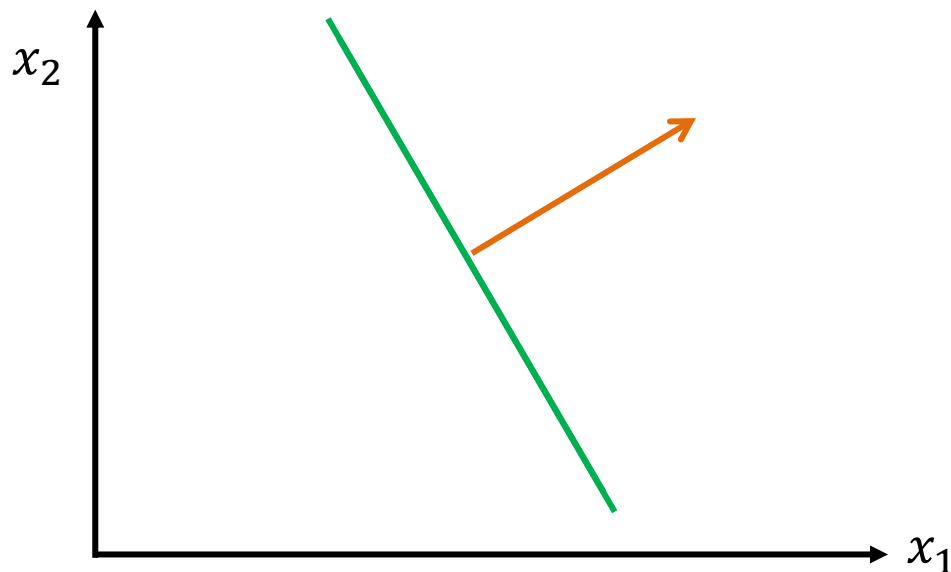
Geometric properties of the dot product

- If the two vectors are orthogonal then $\mathbf{u}'\mathbf{v} = 0$
- If the two vectors are parallel then $\mathbf{u}'\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|$, if they are anti-parallel then $\mathbf{u}'\mathbf{v} = -\|\mathbf{u}\|\|\mathbf{v}\|$
- $\mathbf{u}'\mathbf{u} = \|\mathbf{u}\|^2$, so $\|\mathbf{u}\| = \sqrt{\mathbf{u}'\mathbf{u}} = \sqrt{u_1^2 + \cdots + u_m^2}$ defines the Euclidean vector length



Hyperplanes and normal vectors

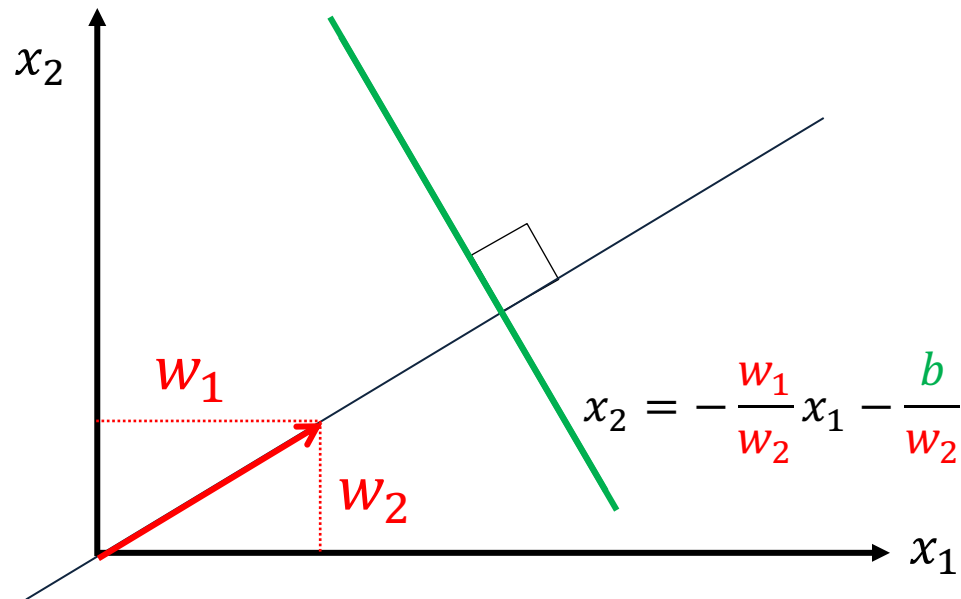
- A hyperplane defined by parameters $\mathbf{w}$ and b is a set of points $\mathbf{x}$ that satisfy $\mathbf{x}'\mathbf{w} + b = 0$
- In 2D, a hyperplane is a line: a line is a set of points that satisfy $w_1x_1 + w_2x_2 + b = 0$



A normal vector for a hyperplane is a vector perpendicular to that hyperplane

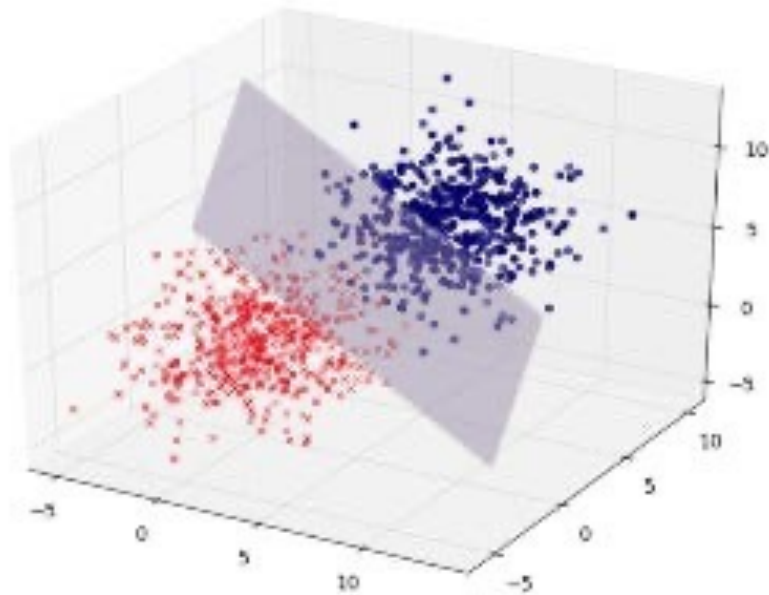
Hyperplanes and normal vectors

- Consider a line defined by w_1 , w_2 and b
- Vector $\mathbf{w} = [w_1, w_2]'$ is a normal vector

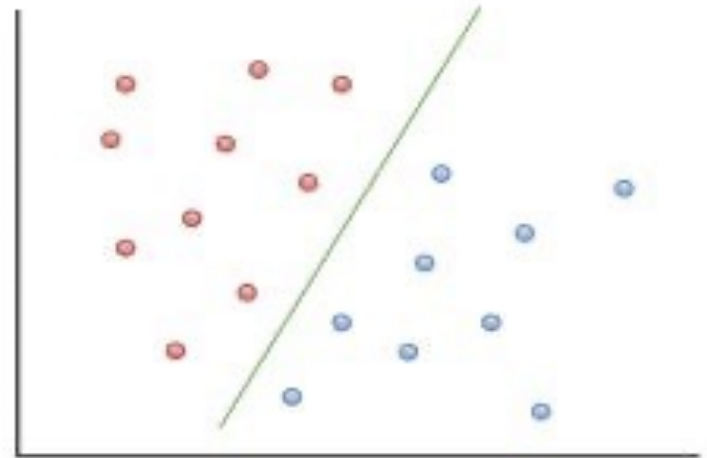


Hyperplane vs line

Hyperplane



Line



L_1 and L_2 norms

- Throughout the subject we will often encounter **norms** that are functions $\mathbb{R}^n \rightarrow \mathbb{R}$ of a particular form
 - * Intuitively, norms measure lengths of vectors in some sense
 - * Often component of objectives or stopping criteria in optimisation-for-ML

- More specifically, we will often use the L_2 norm (*aka* **Euclidean distance**)

$$\|\mathbf{a}\| = \|\mathbf{a}\|_2 \equiv \sqrt{a_1^2 + \cdots + a_n^2}$$

- And also the L_1 norm (*aka* absolute norm or **Manhattan distance**)

$$\|\mathbf{a}\|_1 \equiv |a_1| + \cdots + |a_n|$$

Vector Spaces and Bases

Useful in interpreting matrices and some algorithms like PCA

Linear combinations, Independence

- A **linear combination** of vectors $v_1, \dots, v_k \in V$ some vector space, is a new vector $\sum_{i=1}^k a_i v_i$ for some scalars $a_1, \dots, a_k$
- A set of vectors $\{v_1, \dots, v_k\} \subseteq V$ is called **linearly dependent** if one element v_j can be written as a linear combination of the other elements
- A set that isn't linearly dependent is **linearly independent**
- For formal definition of **vector spaces**:
https://en.wikipedia.org/wiki/Vector_space#Definition

Spans, Bases

- The **span** of vectors $v_1, \dots, v_k \in V$ is the set of all obtainable linear combinations (ranging over all scalar coefficients) of the vectors
- A set of vectors $B = \{v_1, \dots, v_k\} \subseteq V$ is called a **basis** for a vector subspace $V' \subseteq V$ if
 1. The set B is linearly independent; and
 2. Every $v \in V'$ is a linear combination of the set B .
- An **orthonormal basis** is a basis in which:
 1. Each pair of basis vectors are orthogonal (zero dot prod); and
 2. Each vector has norm equal to 1.

Matrices

Some useful facts for ML

Basic matrices

- A rectangular array, often denoted by upper-case, with two indices first for row, second for column
- **Square matrix** has equal dimensions (numbers of rows and columns)
- **Matrix transpose** $\mathbf{A}'$ or $\mathbf{A}^T$ of m by n matrix $\mathbf{A}$ is an n by m matrix with entries $A'_{ij}=A_{ji}$
- A square matrix $\mathbf{A}$ with $\mathbf{A}=\mathbf{A}'$ is called **symmetric**
- The (square) **identity matrix** $\mathbf{I}$ has 1 on the diagonal, 0 off-diagonal
- **Matrix inverse** $\mathbf{A}^{-1}$ of square matrix $\mathbf{A}$ (if it exists) satisfies $\mathbf{A}^{-1}\mathbf{A}=\mathbf{I}$
- See more: [https://en.wikipedia.org/wiki/Matrix_\(mathematics\)](https://en.wikipedia.org/wiki/Matrix_(mathematics))
 - * Including matrix-matrix and matrix-vector products

Matrix eigenspectrum

- Scalar, vector pair $(\lambda, \mathbf{v})$ are called an **eigenvalue-eigenvector** pair of a **square matrix** $\mathbf{A}$ if $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$
 - * Intuition: matrix $\mathbf{A}$ doesn't rotate $\mathbf{v}$ it just **stretches** it
 - * Intuition: the eigenvalue represents stretching factor
- In general eigenvalues may be zero, negative or even complex (imaginary) – we'll only encounter reals

Spectra of common matrices

- Eigenvalues of **symmetric matrices** are always real (no imaginary component)
- A matrix with **linear dependent** columns has some zero eigenvalues (called rank deficient) $\rightarrow$ no matrix inverse exists

Positive (semi)definite matrices

- A symmetric square matrix $\mathbf{A}$ is called positive semidefinite if for all vectors $\mathbf{v}$ we have $\mathbf{v}'\mathbf{A}\mathbf{v} \geq 0$.
 - * Then $\mathbf{A}$ has non-negative eigenvalues
 - * For example, any $\mathbf{A} = \mathbf{X}'\mathbf{X}$ since: $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 \geq 0$
- Further if $\mathbf{v}'\mathbf{A}\mathbf{v} > 0$ holds as a strict inequality then $\mathbf{A}$ is called **positive definite**
 - * Then $\mathbf{A}$ has (strictly) positive eigenvalues

Mini Summary

- Vectors: Vector spaces, dot products, independence, hyperplanes
- Matrices: Eigenvalues, positive semidefinite matrices

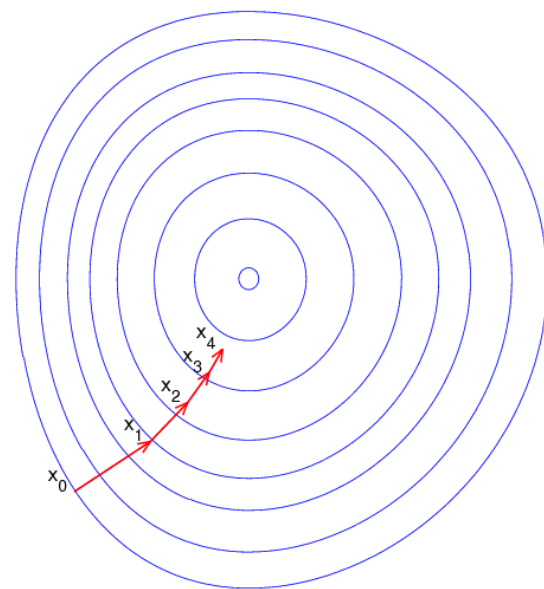
Next: Sequences and limits review/primer

Sequences and Limits

Sequences arise whenever we have iterations (e.g. training loops, growing data sample size). Limits tell us about where sequences tend towards.

Infinite Sequences

- Written like $x_1, x_2, \dots$ or $\{x_i\}_{i \in \mathbb{N}}$
- Formally: a function from the positive (from 1) or non-negative (from 0) integers
- Index set: subscript set e.g. $\mathbb{N}$
- Sequences allow us to reason about test error when training data grows indefinitely, or training error (or a stopping criterion) when training runs arbitrarily long



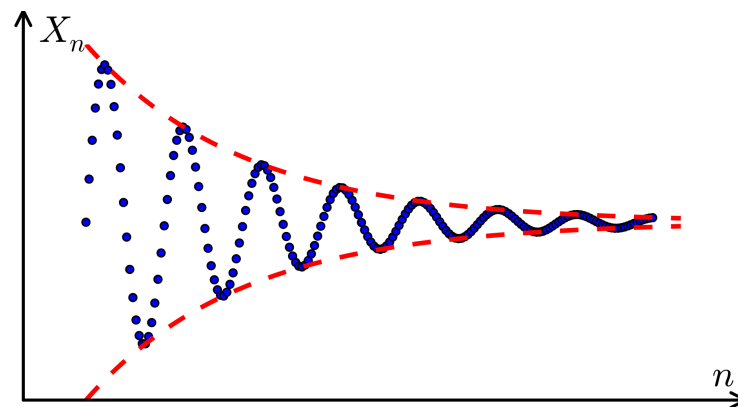
Wikipedia public domain

Limits and Convergence

- A sequence $\{x_i\}_{i \in \mathbb{N}}$ **converges** if its elements become and remain arbitrarily close to a fixed **limit** point L .
- Formally: $x_i \rightarrow L$ if, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $\|x_n - L\| < \varepsilon$

Notes:

- Epsilon ε represents distance of sequence to limit point
- Distance can be arbitrarily small
- Definition says we eventually get that close (at some finite N) and we stay *at least* that close for ever more



Wikipedia public domain

Supremum

Generalising the maximum: When a sequence never quite peaks.

When does the Maximum Exist?

- Can you always take a **max of a set**?
- Finite sets: what's the max of $\{1, 7, 3, 2, 9\}$?
- Closed, bounded intervals: what's the max of $[0,1]$?
- Open, bounded intervals: what's the max of $[0,1)$?
- Open, unbounded intervals: what's the max of $[0,\infty)$?

What about “Least Upper Bound”?

- Can you always take a **least-upper-bound of a set**? (much more often!)
- Finite sets: what's the max of $\{1, 7, 3, 2, 9\}$?

max=9 LUB=9

- Closed, bounded intervals: what's the max of $[0,1]$?

max=1 LUB=1

- Open, bounded intervals: what's the max of $[0,1)$?

max=N/A LUB=1

- Open, unbounded intervals: what's the max of $[0,\infty)$?

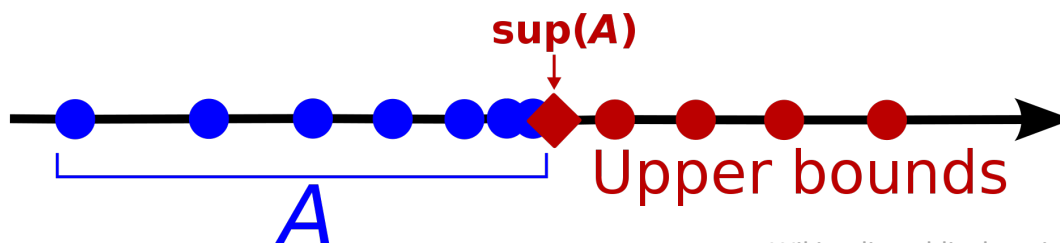
max=N/A LUB= ∞

The Supremum

- Consider any subset S of the reals
- Upper bound** $u \in \mathbb{R}^+$ of set S has: $u \geq x$ for all $x \in S$
- If u is no bigger than any other upper bound of S then it's called a least upper bound or **supremum** of S , written as $\sup(S)$ and pronounced “soup”:
 - * $z \geq u$ for all upper bounds $z \in \mathbb{R}^+$ of S
- When we don't know, or can't guarantee, that a set or sequence has a max, it is better to use its sup



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Infimum

- The greatest lower bound or **infimum** is generalisation of the minimum
- Written $\inf(S)$ pronounced “inf”
- Useful if we’re minimising training error but don’t know if the minimum is ever attained.

Mini Summary

- Sequences
- Limits of sequences
- Supremum is the new maximum

Next time: L02 Statistical schools

Jupyter notebooks setup and launch (at home)